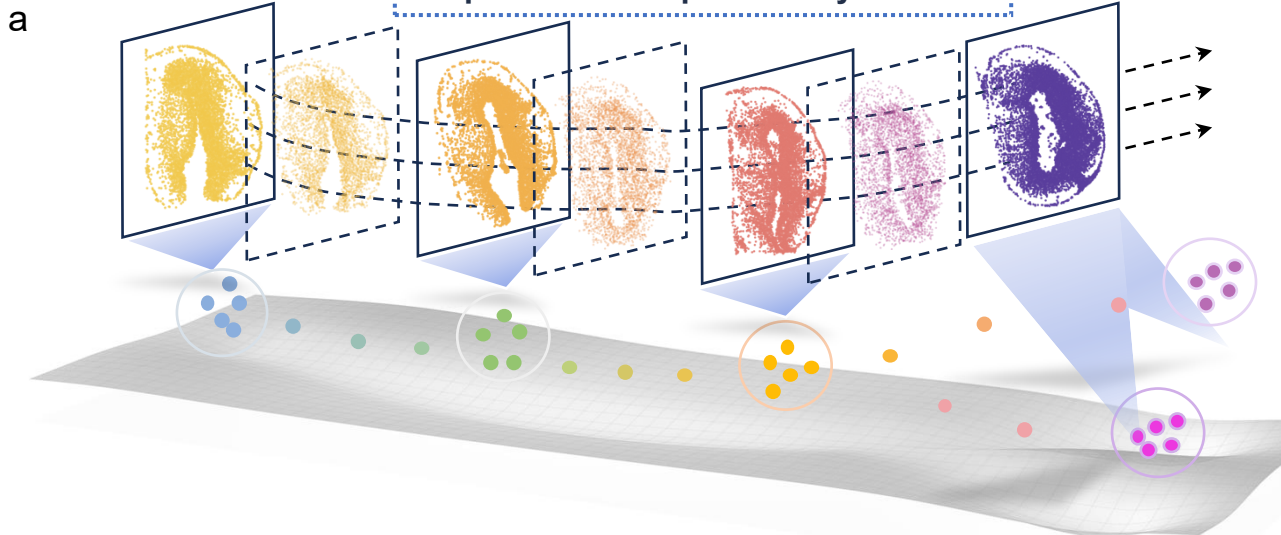


Spatial Transcriptomics Dynamics

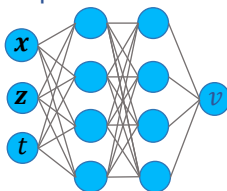


Training SpaPOT Model

Spatiotemporal potential-transport equation

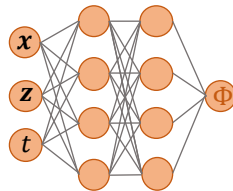
$$\partial_t \rho + \nabla_x \cdot (\rho \mathbf{v}) - \nabla_z \cdot (\rho \nabla_z \Phi) = g \rho$$

Spatial field $\mathbf{v}$



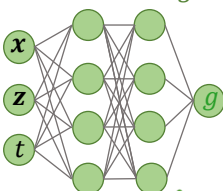
Velocity in space

Potential field Φ



Potential landscape

Growth field g



Cell growth/decay

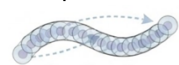
Optimization objective

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{mat}} + \lambda_2 \mathcal{L}_{\text{Spa}} + \lambda_3 \mathcal{L}_{\text{phy}} + \lambda_4 \mathcal{L}_{\text{dyn}}$$

Matching loss



Spatial loss



Physical loss

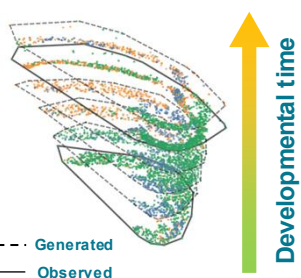


Dynamic loss

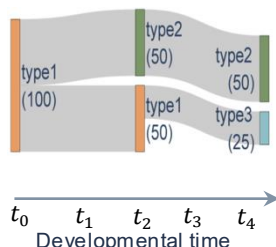


Downstream Analyses and Applications

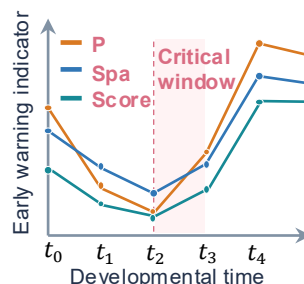
1 Spatial slice generation



2 Lineage inference



3 Critical point warning



4 Perturbed gene program

